Gradient-Guided Density Peak Clustering

Yikun Zhang[1,2]

Joint work with *Professor Yen-Chi Chen*[3]

[1]*Department of Statistics, University of Chicago*
[2]*NSF-Simons AI Institute for the Sky (SkAI Institute)*
[3]*Department of Statistics, University of Washington*

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Background and Introduction

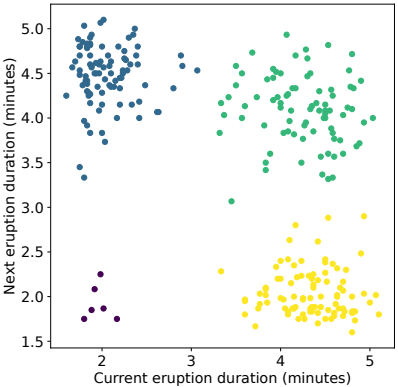


Clustering: a data analysis technique that partitions a set of objects into groups (called **clusters**) based on their similarity.

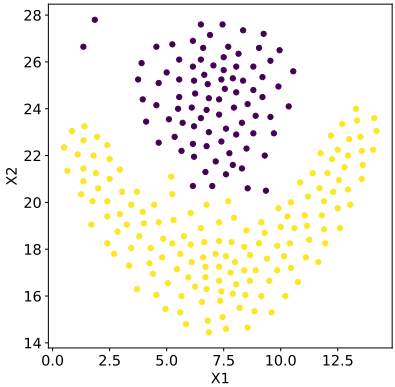
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Old Faithful dataset (Azzalini and Bowman, 1990).



Flame dataset (Fu and Medico, 2007).

What is Density Peak Clustering (DPC)?

1. DPC (Rodriguez and Laio, 2014) is a type of density-based clustering:

$$\text{Partition } \mathbb{X}_n = \{\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n\} \quad \overset{\text{based on}}{\Longrightarrow} \quad \text{density function } p : \mathbb{R}^d \to \mathbb{R} \text{ of } \mathbb{X}_n.$$

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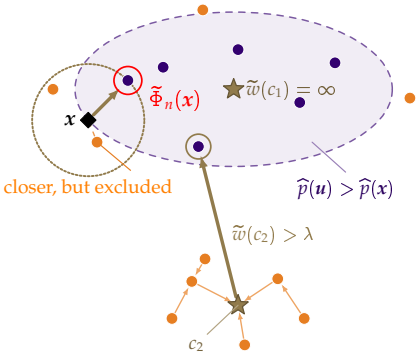
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2. Given $\mathbb{X}_n = \{\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n\}$ and any $\boldsymbol{x} \in \mathbb{R}^d$, define
   - 1NN with higher density:

$$\widetilde{\Phi}_n(\boldsymbol{x}) = \underset{\boldsymbol{X}_i \in \mathbb{X}_n}{\arg\min} \{||\boldsymbol{X}_i - \boldsymbol{x}|| : \widehat{p}(\boldsymbol{X}_i) > \widehat{p}(\boldsymbol{x})\}.$$

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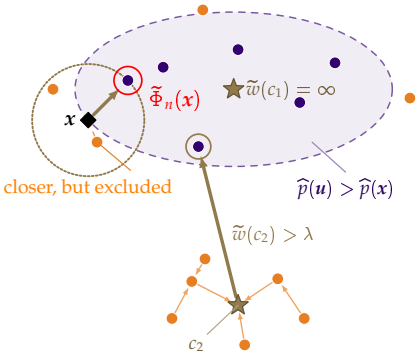
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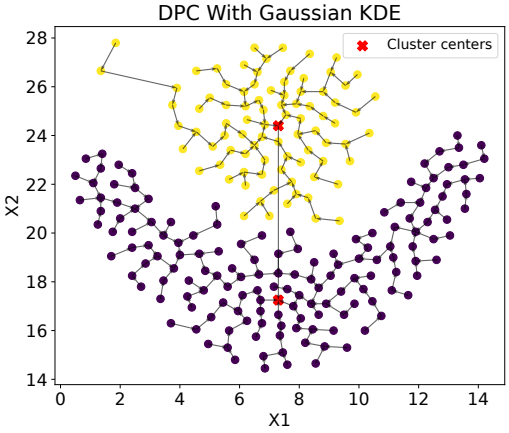
3. Identify the cluster centers

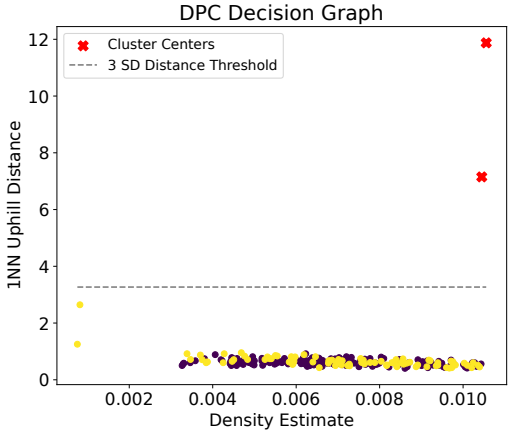
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Advantages of DPC

Question: Why do we study or even use DPC?

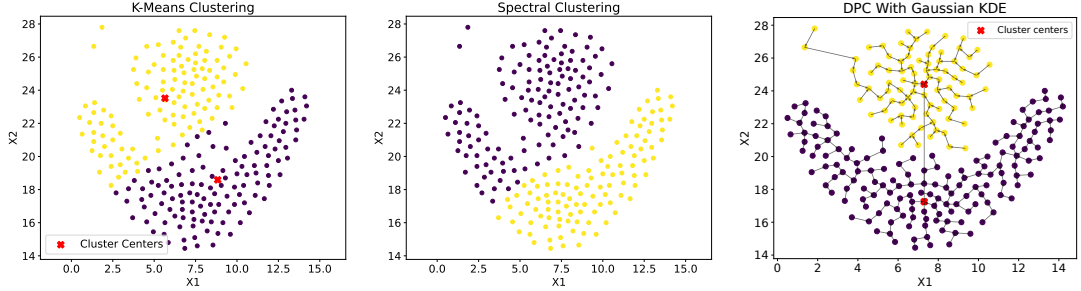
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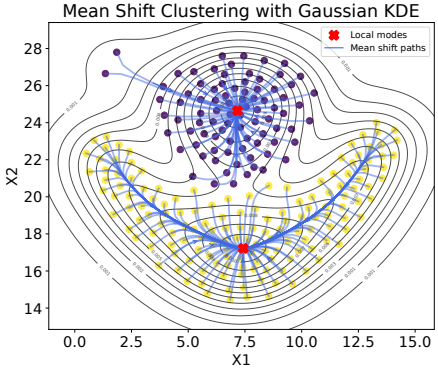


2. More adaptive to irregular cluster shapes.

The gradient ascent flow of p starting at $\boldsymbol{x}$ is defined as

$$\boldsymbol{\gamma}_{\boldsymbol{x}}'(t) = \nabla p(\boldsymbol{\gamma}_{\boldsymbol{x}}(t)), \qquad \boldsymbol{\gamma}_{\boldsymbol{x}}(0) = \boldsymbol{x}.$$

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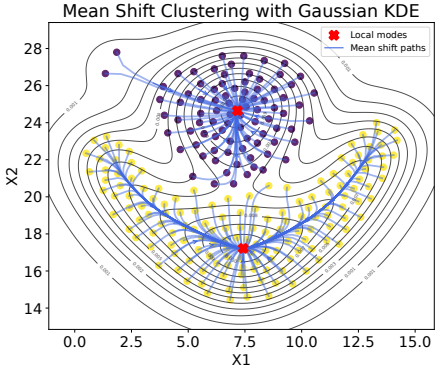


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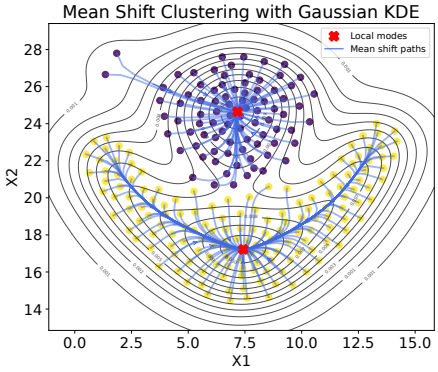
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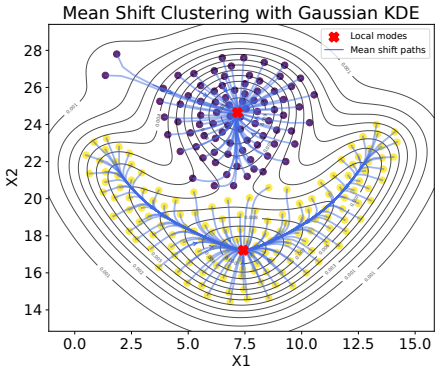
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Drawback of Mean Shift Clustering:

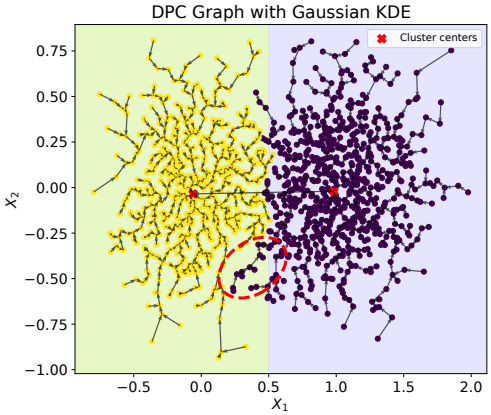
- From *every* $\boldsymbol{X}_i$, it *repeatedly* estimates/evaluates the gradient until convergence.

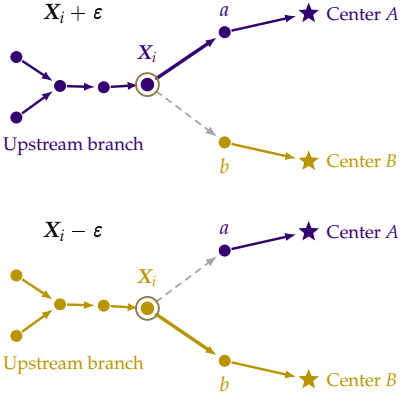
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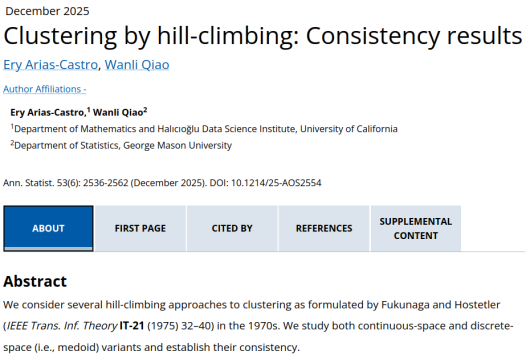
A single perturbation redirects the whole upstream branch (Xie et al., 2016; Seyedi et al., 2019).

Open Problem: Pathwise Consistency of DPC

Domino effect obscures a *population counterpart* of the DPC path from $\boldsymbol{X}_i \in \mathbb{X}_n$ to a cluster center in $\widetilde{\mathcal{M}}_\lambda$.

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Clustering by hill-climbing: Consistency results

Ery Arias-Castro, Wanli Qiao

Author Affiliations -

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Abstract

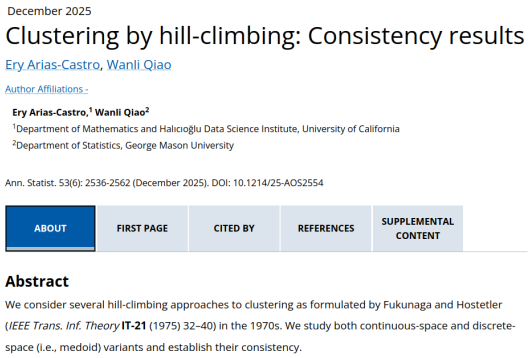
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Section 5.4.2: "A population analog of quick shift† is not straightforward to define ... As far as we know, proving (or disproving) the consistency of quick shift over the entire basins of attraction remains an open problem."

†Quick shift (Vedaldi and Soatto, 2008) is essentially the same algorithm as DPC.

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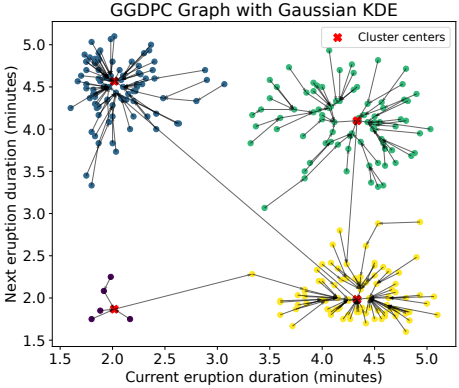
Goal: Resolve the instability of DPC with a minimal fix and study its (pathwise) stability theory.

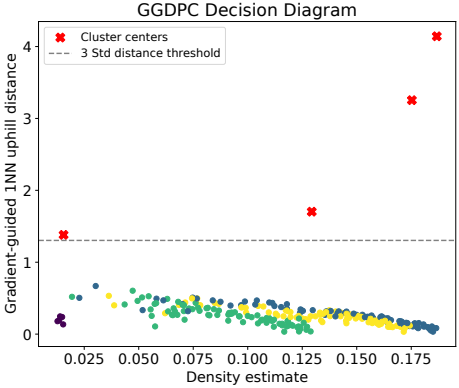
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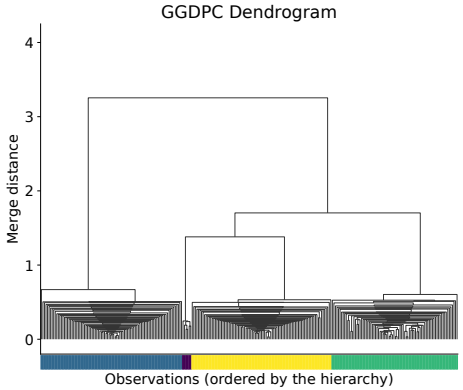


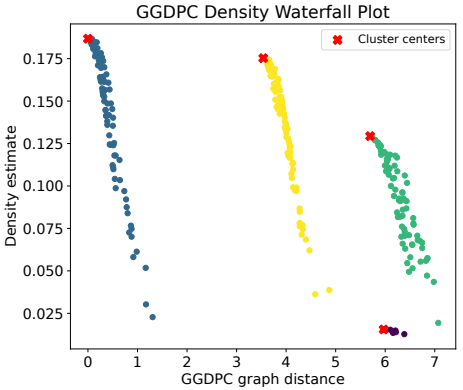


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3. ***"Stable Clustering":***

   GGDPC ascending paths $\approx$ Gradient ascent flow of p as $n \to \infty$.

Gradient-Guided Density Peak Clustering (GGDPC)



Let $\widehat{p}$ and $\widehat{g}$ be estimators of p and its gradient ∇p. Given $\mathbb{X}_n = \{\boldsymbol{X}_1, ..., \boldsymbol{X}_n\}$,

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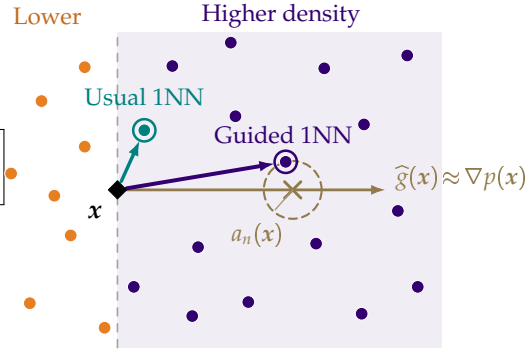
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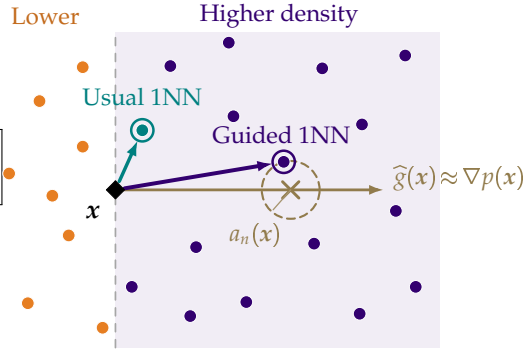
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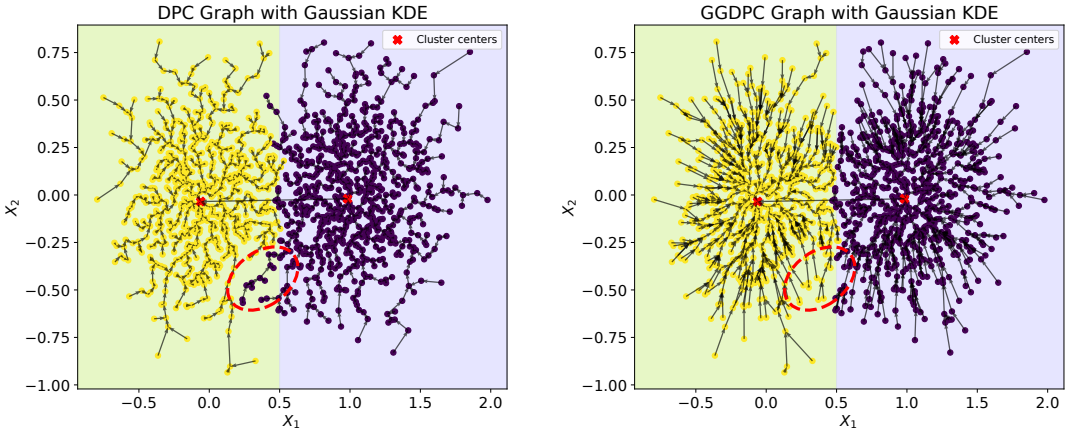
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- Density is compared with the *original* point.
- $\widehat{\mathcal{M}}_\lambda = \{\boldsymbol{X}_i \in \mathbb{X}_n : \widehat{w}_n(\boldsymbol{X}_i) > \lambda\}$ for $\lambda > 0$.

Gaussian mixture: $0.5 \cdot \mathcal{N}(\boldsymbol{\mu}_1, 0.09\boldsymbol{I}_2) + 0.5 \cdot \mathcal{N}(\boldsymbol{\mu}_2, 0.09\boldsymbol{I}_2)$, $\boldsymbol{\mu}_1 = (0,0)^T$, $\boldsymbol{\mu}_2 = (1,0)^T$.

- $n = 1500$ and population separatrix is $x_1 = 0.5$.



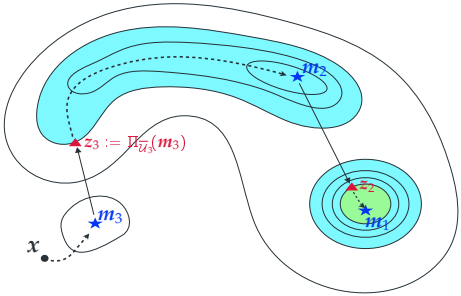
On the same sample, the GGDPC graph follows the modal basins better.

Convergence of GGDPC Clustering



Question: Is $\widehat{\mathcal{M}}_\lambda = \{\boldsymbol{X}_i \in \mathbb{X}_n : \widehat{w}_n(\boldsymbol{X}_i) > \lambda\}$ statistically consistent?

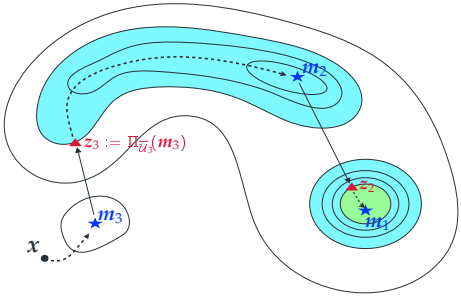
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Assume p is Morse with a compact support.

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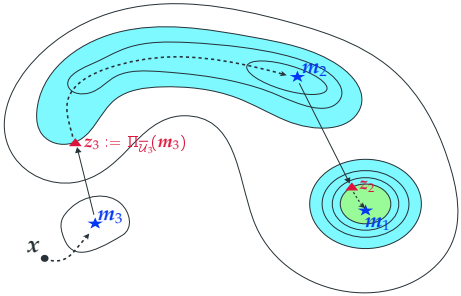
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$$\text{Haus}(\widehat{\mathcal{M}}_\lambda, \mathcal{M}_\lambda) = \max\left\{\sup_{\boldsymbol{x} \in \widehat{\mathcal{M}}_\lambda} d(\boldsymbol{x}, \mathcal{M}_\lambda), \sup_{\boldsymbol{y} \in \mathcal{M}_\lambda} d(\boldsymbol{y}, \widehat{\mathcal{M}}_\lambda)\right\}.$$

Theorem (Zhang and Chen 2026; see also Jiang 2017; Verdinelli and Wasserman 2018)

1. *p is a Morse function supported on a compact regular set $\mathcal{C}$, and* $\inf_{x \in \mathcal{C}} p(\boldsymbol{x}) \geq p_{\min} > 0$.
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For a smooth estimator $\widehat{p}$ *with* $\widehat{g} = \nabla \widehat{p}$,

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- Minimaxity when $d \leq 6$ (Romano, 1988; Arias-Castro et al., 2022):

$$||\widehat{g} - \nabla p||_\infty = O_P\left(\left(\frac{\log n}{n}\right)^{2/(d+6)}\right) \implies \text{Haus}\left(\mathcal{M}_\lambda, \widehat{\mathcal{M}}_\lambda\right) = O_P\left(\left(\frac{\log n}{n}\right)^{\min\{2/(d+6), 1/d\}}\right).$$

Convergence of Adjusted Rand Index (ARI)

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- Population basins of attraction $\mathcal{C}_1, ..., \mathcal{C}_M$ of p:

$$\mathcal{C}_j = \left\{ \boldsymbol{x} \in \mathcal{C} : \lim_{t \to \infty} \boldsymbol{\gamma}_{\boldsymbol{x}}(t) = \boldsymbol{m}_j \right\}, \qquad \boldsymbol{\gamma}'_{\boldsymbol{x}}(t) = \nabla p(\boldsymbol{\gamma}_{\boldsymbol{x}}(t)) \quad \text{and} \quad \boldsymbol{\gamma}_{\boldsymbol{x}}(0) = \boldsymbol{x}.$$

- Define the population modal partition of $\mathbb{X}_n = \{\boldsymbol{X}_1, ..., \boldsymbol{X}_n\}$ as

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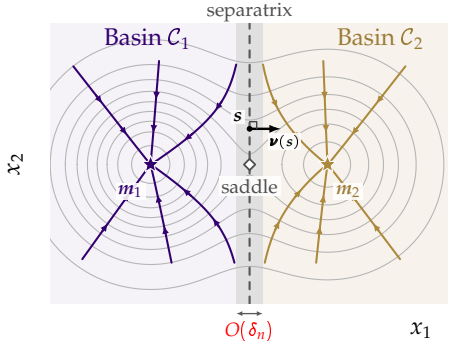
ARI (Hubert and Arabie, 1985): $\text{ARI}\left(\widehat{\mathcal{P}}_{n,\lambda}, \mathcal{P}_n^*\right) = \frac{\text{RI} - \mathbb{E}_0(\text{RI})}{1 - \mathbb{E}_0(\text{RI})}$ with $\text{RI} = \frac{TP+TN}{\binom{n}{2}}$.

- $\text{ARI} = 1$: identical clusterings; $\mathbb{E}_0[\text{ARI}] = 0$: random labeling.

Normal Repulsion Condition: For $s \in \mathrm{Reg}(\partial \mathcal{C}_j)$,

$$\exists \rho_{\mathcal{S}} > 0 \quad \text{s.t.} \quad \nu(s)^T \nabla^2 p(s) \nu(s) \geq \rho_{\mathcal{S}},$$

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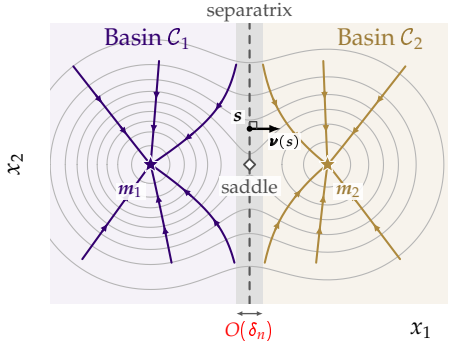


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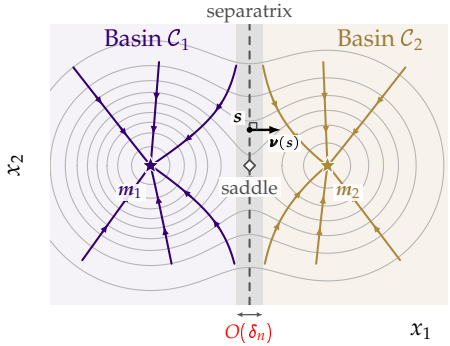


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Theorem (Zhang and Chen 2026)

1. *Morse function p + normal repulsion + smooth estimator $\widehat{p}$ with $\widehat{g} = \nabla \widehat{p}$.*
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$$1 - \mathrm{ARI}\left(\widehat{\mathcal{P}}_{n,\lambda}, \mathcal{P}_n^*\right) = O_P\left(\eta_n + \frac{q_n}{\eta_n} + ||\widehat{g} - \nabla p||_\infty\right), \quad \forall \lambda \in \left(0, \min_{2 \leq j \leq |\mathcal{M}|} \psi_j\right).$$

Convergence Rates: Modal Consistency vs. ARI

Let $r_n = \frac{\log n}{n}$, $q_n = r_n^{\frac{1}{d}}$, and $e_n = ||\widehat{g} - \nabla p||_\infty$.

Criterion	Step size	Error bound
Modal consistency (Hausdorff)	$\eta_n = o(1)$	$O_P(q_n + e_n)$
Clustering $(1 - \text{ARI})$	$\eta_n \asymp \sqrt{q_n}$	$O_P(\sqrt{q_n} + e_n)$

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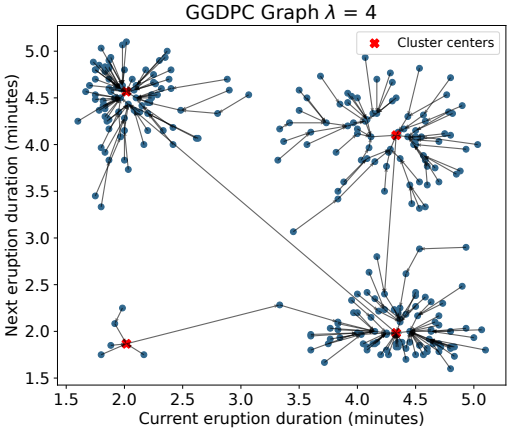
For (boundary-corrected) KDE $\widehat{p}_n$ with $e_n = O_P\left(r_n^{\frac{2}{d+6}}\right)$:

Criterion	Rate	Interpretation
Modal consistency (Hausdorff)	$O_P\left(r_n^{\min\left\{\frac{2}{d+6}, \frac{1}{d}\right\}}\right)$	Minimax up to logs for $d \leq 6$
Clustering $(1 - \text{ARI})$	$O_P\left(r_n^{\min\left\{\frac{2}{d+6}, \frac{1}{2d}\right\}}\right)$	1NN error dominates for $d \geq 3$

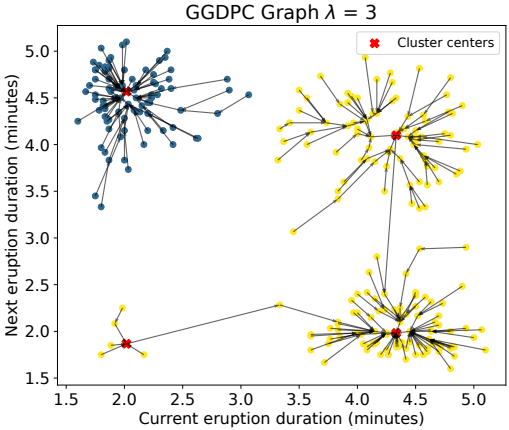
GGDPC Dendrogram



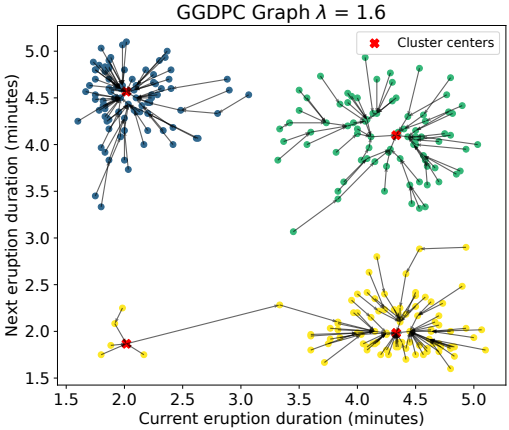
G_λ: trim the outgoing edge from *every* $\boldsymbol{X}_i$ with $\widehat{w}_n(\boldsymbol{X}_i) > \lambda$ in the GGDPC graph G.



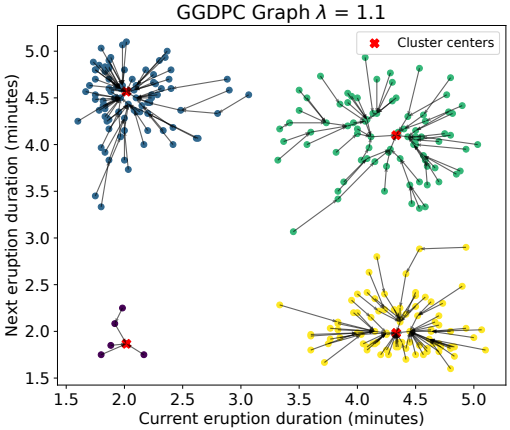
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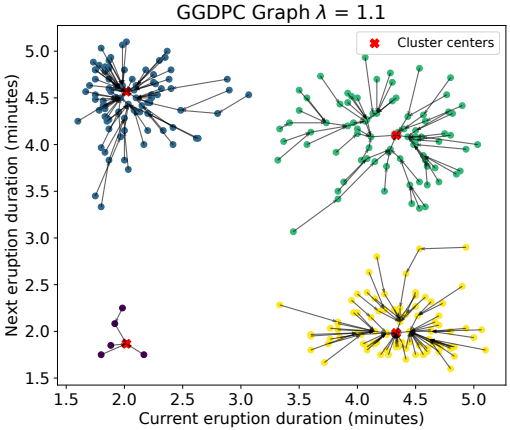


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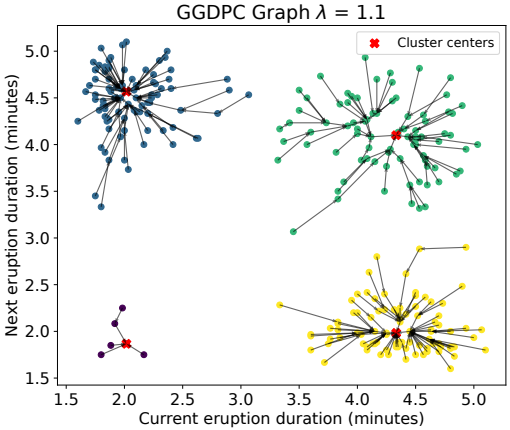
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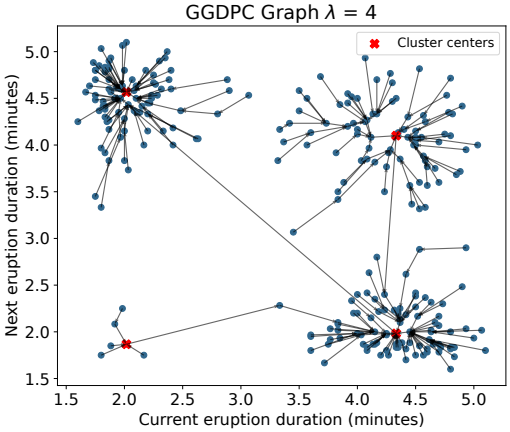


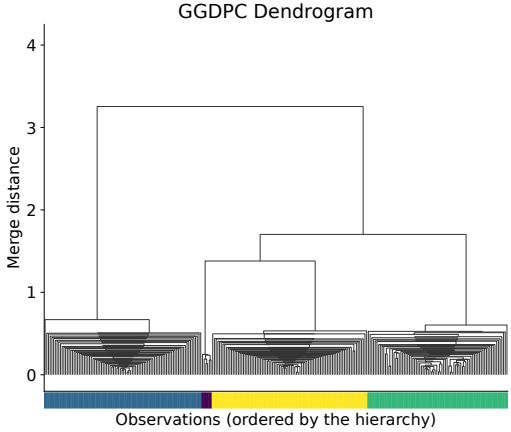
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Hierarchy of $\mathcal{T}_G$:

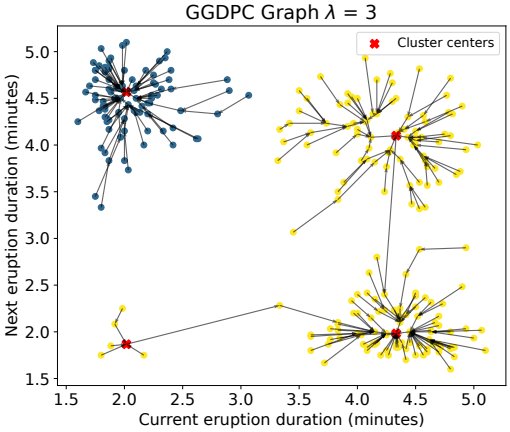
- For any $0 \leq \lambda \leq \lambda' < \infty$, the partition $\mathcal{T}_G(\lambda)$ refines $\mathcal{T}_G(\lambda')$.

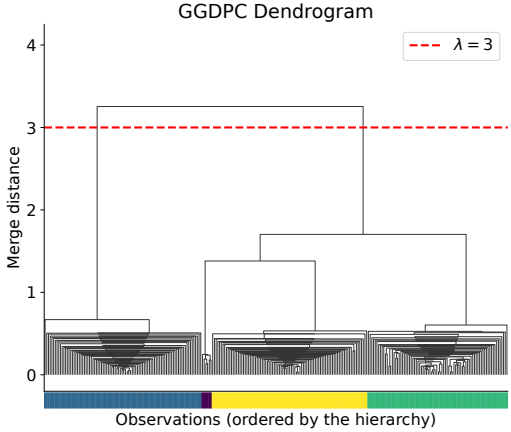
Divisive Approach: Start from G at $\lambda \geq \max \widehat{w}_n(\boldsymbol{X}_i)$ and remove edges as $\lambda \downarrow 0$.



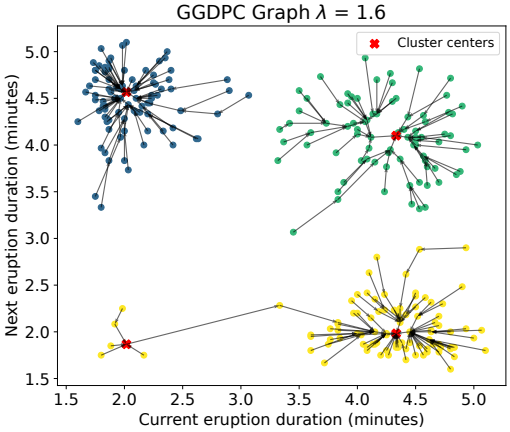


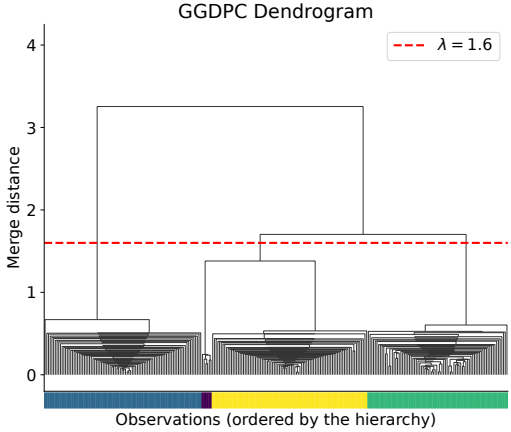
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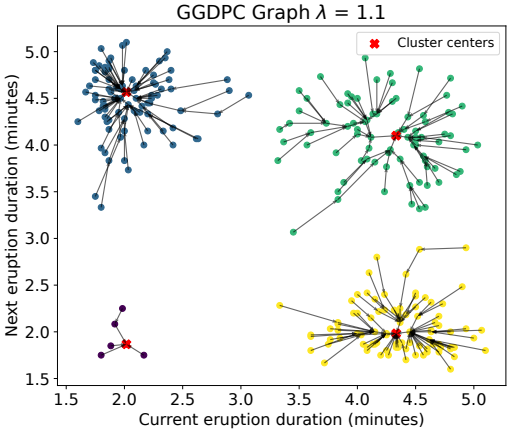


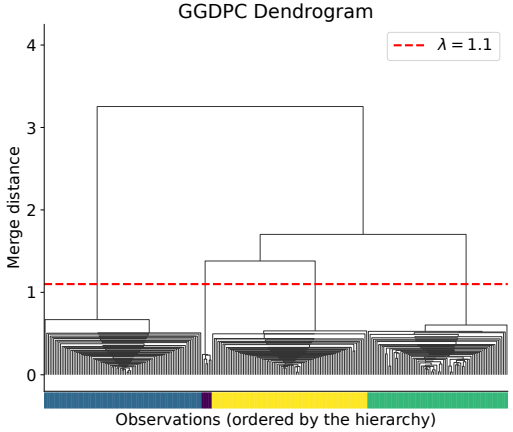
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Agglomerative Approach: Choose $d_\star > d_{\max} = \max_{1 \leq i,j \leq n} \left|\left| \boldsymbol{X}_i - \boldsymbol{X}_j \right|\right|$ and define a *symmetric* matrix $D \in \mathbb{R}^{n \times n}$ with $D_{ii} = 0$ and

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- Apply the single linkage clustering to D.

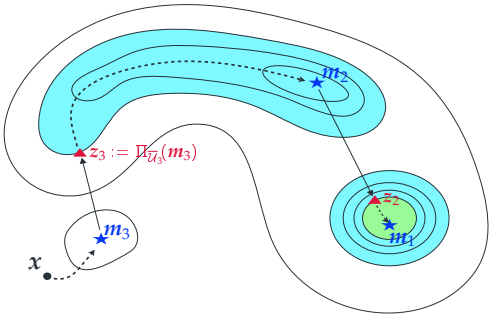
Proposition (Zhang and Chen 2026)

Let $\mathcal{T}_{\mathrm{SL}}(\lambda)$ be the partition of $\mathbb{X}_n$ by cutting the single linkage dendrogram induced by D at height λ. Then,

$$\mathcal{T}_{\mathrm{SL}}(\lambda) = \mathcal{T}_G(\lambda) \qquad \textit{for every} \quad \lambda \geq 0.$$

Question: Does the GGDPC dendrogram converge to a population limit as $n \to \infty$?

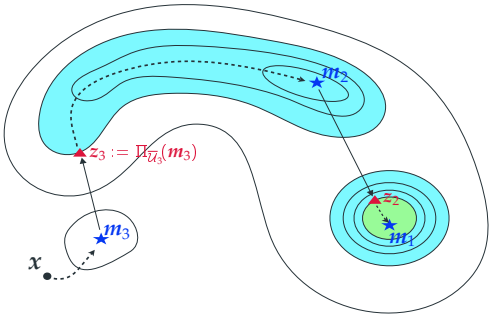
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$$\overline{\mathcal{U}}_j := \overline{\left\{ x \in \mathbb{R}^d : p(x) > p(m_j) \right\}} \text{ and}$$

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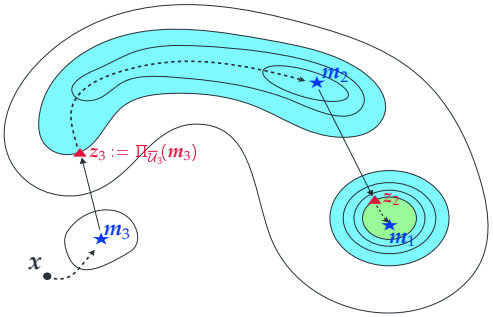
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Define a weighted graph G_{mode} on $\mathcal{M}$:

- edges: $\boldsymbol{m}_j \to \boldsymbol{m}_{\pi(j)}, j = 2, \ldots, M$ such that $\boldsymbol{z}_j \in \mathcal{C}_{\pi(j)}$ (basin of attraction of $\boldsymbol{m}_{\pi(j)}$);
- $\boldsymbol{m}_j \to \boldsymbol{m}_{\pi(j)}$ has weights $w(\boldsymbol{m}_j)$.

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$$z_j := \arg\min_{x \in \overline{\mathcal{U}}_j} w(m_j), \quad w(m_j) = d(m_j, \overline{\mathcal{U}}_j).$$

Define a weighted graph G_{mode} on $\mathcal{M}$:

- edges: $m_j \to m_{\pi(j)}, j = 2, ..., M$ such that $z_j \in \mathcal{C}_{\pi(j)}$ (basin of attraction of $m_{\pi(j)}$);
- $m_j \to m_{\pi(j)}$ has weights $w(m_j)$.

Modal Distance Dendrogram: define a mapping $\mathcal{T}_{\text{mode}} : [0, \infty) \to \mathcal{P}(\mathcal{M})$.

- $\mathcal{P}(\mathcal{M})$: the collection of all partitions of $\mathcal{M}$.

Dendrograms $\Longleftrightarrow$ **Ultrametric**[1] **on** $\mathbb{X}_n$ (Carlsson and Mémoli, 2010):

$$\begin{aligned} u(\boldsymbol{X}_i, \boldsymbol{X}_j) &= \min \left\{ \lambda \geq 0 : \boldsymbol{X}_i, \boldsymbol{X}_j \text{ belong to the same block of } \mathcal{T}_G(\lambda) \right\} \\ &= \max \left\{ \widehat{w}_n(\boldsymbol{X}_k) : \left(\boldsymbol{X}_k \to \widehat{\Phi}_n(\boldsymbol{X}_k) \right) \in \mathrm{Path}_G(\boldsymbol{X}_i, \boldsymbol{X}_j) \right\}. \end{aligned}$$

- $\mathrm{Path}_G(\boldsymbol{X}_i, \boldsymbol{X}_j)$: the edge set in G joining $\boldsymbol{X}_i$ and $\boldsymbol{X}_j$.

[1] An ultrametric $u : \mathbb{X} \times \mathbb{X} \to \mathbb{R}_+$ is a valid metric and satisfies $\max\{u(\boldsymbol{x}, \boldsymbol{z}), u(\boldsymbol{z}, \boldsymbol{y})\} \geq u(\boldsymbol{x}, \boldsymbol{y})$ for any $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{X}$.

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Similarly, $u_{\mathcal{M}}(\boldsymbol{m}_j, \boldsymbol{m}_k) = \max\left\{w(\boldsymbol{m}_\ell) : \left(\boldsymbol{m}_\ell \to \boldsymbol{m}_{\pi(\ell)}\right) \in \mathrm{Path}_{G_{\mathcal{M}}}(\boldsymbol{m}_j, \boldsymbol{m}_k)\right\}$.

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Theorem (Gromov-Hausdorff convergence; Zhang and Chen 2026)

1. *Morse function p + smooth estimator $\widehat{p}$ + no flat tangential region at $\boldsymbol{z}_j$.*
2. $\eta_n + \frac{q_n a_n}{\eta_n} = o(1)$, $||\widehat{p} - p||_\infty = o_P(1)$, $||\widehat{g} - \nabla p||_\infty = o_P(1)$, $\left|\left|\nabla^2 \widehat{p} - \nabla^2 p\right|\right|_\infty = o_P(1)$.

$$d_{\mathrm{GH}}((\mathcal{T}_G, \mathbb{X}_n), (\mathcal{T}_{\mathrm{mode}}, \mathcal{M})) = O_P\left(\eta_n + a_n\left[\frac{q_n}{\eta_n} + ||\widehat{g} - \nabla p||_\infty\right]\right) \quad \textit{for any } a_n \to \infty.$$

[1] An ultrametric $u : \mathbb{X} \times \mathbb{X} \to \mathbb{R}_+$ is a valid metric and satisfies $\max\{u(x, z), u(z, y)\} \geq u(x, y)$ for any $x, y, z \in \mathbb{X}$.

Convergence Rate Comparisons

Let $r_n = \frac{\log n}{n}$, $q_n = r_n^{1/d}$, $e_n = ||\widehat{g} - \nabla p||_\infty$, and choose any deterministic sequence $a_n \to \infty$ slowly so that $a_n q_n = o(1)$ and $a_n e_n = o_P(1)$.

Criterion	Step size	Error bound
Modal consistency (Hausdorff)	$\eta_n = o(1)$	$O_P(q_n + e_n)$
Clustering ($1 - \text{ARI}$)	$\eta_n \asymp \sqrt{q_n}$	$O_P(\sqrt{q_n} + e_n)$
Dendrogram (d_{GH})	$\eta_n \asymp \sqrt{q_n}$	$O_P(\sqrt{a_n q_n} + a_n e_n)$

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For (boundary-corrected) KDE $\widehat{p}_n$ with $e_n = O_P\left(r_n^{2/(d+6)}\right)$:

Criterion	Rate	Interpretation
Modal consistency (Hausdorff)	$O_P\left(r_n^{\min\left\{\frac{2}{d+6}, \frac{1}{d}\right\}}\right)$	Minimax up to logs for $d \leq 6$
Clustering $(1 - \text{ARI})$	$O_P\left(r_n^{\min\left\{\frac{2}{d+6}, \frac{1}{2d}\right\}}\right)$	1NN error dominates for $d \geq 3$
Dendrogram (d_{GH})	$O_P\left((\log\log n) r_n^{\min\left\{\frac{2}{d+6}, \frac{1}{2d}\right\}}\right)$	Same polynomial rate as ARI

Stability of GGDPC Paths and Density Waterfall



GGDPC uphill path from $\widehat{\boldsymbol{Y}}_0 = \boldsymbol{x} \in \mathcal{C}_j$ to $\widehat{\boldsymbol{Y}}_{k+1} = \widehat{\Phi}_n(\widehat{\boldsymbol{Y}}_k), k = 1, ..., \widehat{T}_n$:

$$\widehat{L}_n(\boldsymbol{x}) := \sum_{k=0}^{\widehat{T}_n - 1} \left|\left| \widehat{\boldsymbol{Y}}_{k+1} - \widehat{\boldsymbol{Y}}_k \right|\right|.$$

- $\widehat{\Phi}_n(\boldsymbol{x}) = \arg\min_{\boldsymbol{X}_i \in \mathbb{X}_n} \{||\boldsymbol{X}_i - [\boldsymbol{x} + \eta_n \widehat{g}(\boldsymbol{x})]|| : \widehat{p}(\boldsymbol{X}_i) > \widehat{p}(\boldsymbol{x})\}$.
- $\widehat{T}_n := \inf\left\{k \geq 0 : \widehat{\Phi}_n(\widehat{\boldsymbol{Y}}_k) \text{ is undefined or does not belong to } \mathcal{C}_j\right\}$.

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Gradient ascent flow of p from $\boldsymbol{\gamma}_{\boldsymbol{x}}(0) = \boldsymbol{x}$ is $\boldsymbol{\gamma}'_{\boldsymbol{x}}(t) = \nabla p(\boldsymbol{\gamma}_{\boldsymbol{x}}(t))$:

$$L(\boldsymbol{x}) = \int_0^\infty ||\nabla p(\boldsymbol{\gamma}_{\boldsymbol{x}}(t))|| \, dt.$$

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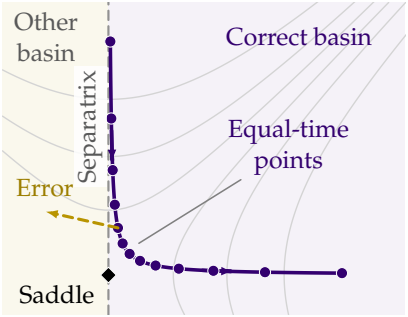
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Section 5.4.2 in Arias-Castro and Qiao (2025): "... proving (or disproving) the consistency of quick shift over the entire basins of attraction remains an open problem."

Question: Does $\sup_{\boldsymbol{x} \in \mathcal{C}_j} \left|\widehat{L}_n(\boldsymbol{x}) - L(\boldsymbol{x})\right| \xrightarrow{P} 0$, and at what rate?

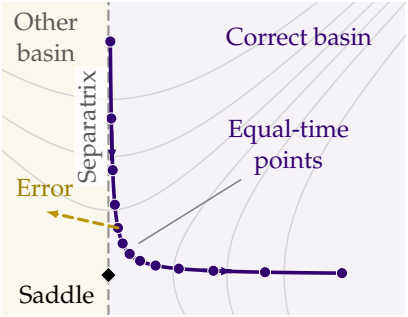
Near a boundary saddle



1NN and gradient errors may push GGDPC paths into the wrong basin.

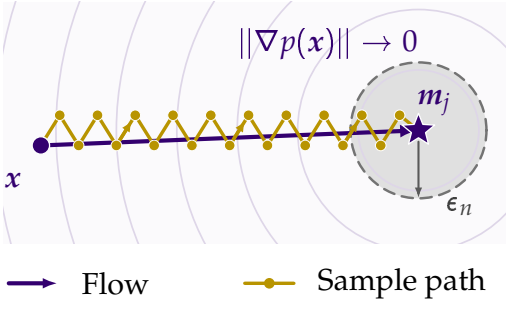
Challenges in Proving Path-Length Consistency

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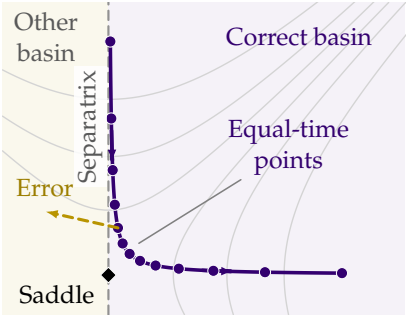
Near a local mode



1NN dominates gradient ascent, and sample paths can zigzag.

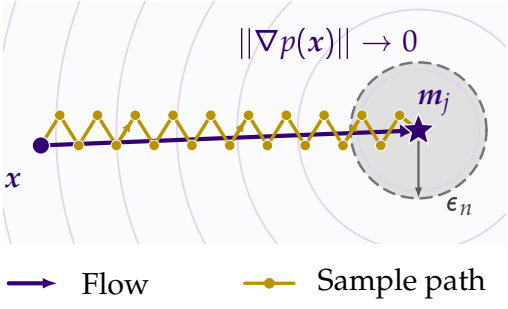
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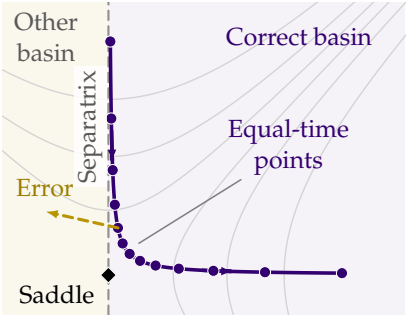
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1. Besides normal repulsion, assume ∇p is C^1-linearizable near boundary saddles $\boldsymbol{s}$:

$$\boldsymbol{\gamma}_{\boldsymbol{x}}'(t) = \nabla p(\boldsymbol{\gamma}_{\boldsymbol{x}}(t)) \quad \overset{\text{smooth map } \Phi:\ \boldsymbol{y}(t)=\Phi(\boldsymbol{\gamma}_{\boldsymbol{x}}(t))}{\Longleftrightarrow} \quad \boldsymbol{y}'(t) = A\boldsymbol{y}(t), \quad A = \nabla^2 p(\boldsymbol{s}) \in \mathbb{R}^{d\times d}.$$

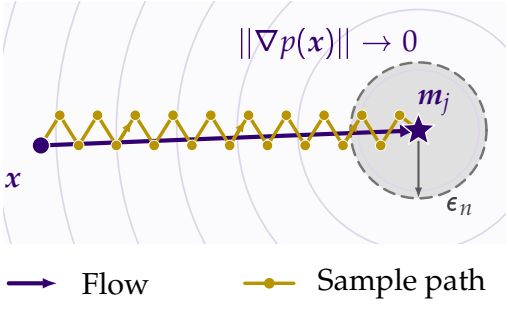
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2. In a shrinking neighborhood $B(\boldsymbol{m}_j, \epsilon_n)$, the total path length is $O_P(n\epsilon_n^d \cdot \epsilon_n)$.

Theorem (Zhang and Chen 2026)

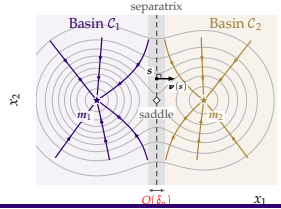
1. *Morse function p + normal repulsion + smooth $\widehat{p}$ + C^1-linearization on boundary saddles.*
2. *For a deterministic sequence $\delta_n \downarrow 0$, assume*
   - $\eta_n + \frac{q_n}{\eta_n} = o(\delta_n),\ \delta_n \left|\log\left(\frac{q_n}{\eta_n}\right)\right| = O(1),\ \frac{q_n \log n}{\eta_n^{d+1}} = o(1)$ *with* $q_n := \left(\frac{\log n}{n}\right)^{1/d}$;
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Then, for some constant $c > 0$,

$$\sup_{x \in \mathcal{C}_j : d(x, \partial \mathcal{C}_j) > c\,\delta_n} \left|\widehat{L}_n(x) - L(x)\right| = O_P\left(\frac{q_n}{\eta_n}\left|\log\left(\frac{q_n}{\eta_n}\right)\right| + \frac{q_n}{\eta_n \delta_n} + \frac{\eta_n}{\delta_n} + \frac{q_n \log n}{\eta_n^{d+1}} + \frac{||\widehat{g} - \nabla p||_\infty}{\delta_n}\right).$$



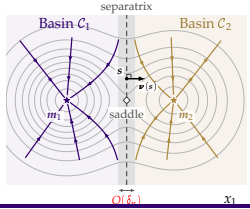
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- Cyan: error rate away from local modes and boundary saddles.
- Blue: error rate near boundary saddles.
- Orange: Choose $\epsilon_n \asymp \frac{q_n}{\eta_n}$ so that $n\epsilon_n^{d+1} \asymp \frac{q_n \log n}{\eta_n^{d+1}}$.
- Green: gradient estimation error (near boundary saddles).



Let $r_n = \frac{\log n}{n}$, $q_n = r_n^{1/d}$, $e_n = ||\widehat{g} - \nabla p||_\infty$, and choose any deterministic sequence $a_n \to \infty$ slowly so that $a_n q_n = o(1)$ and $a_n e_n = o_P(1)$.

Path-Length Consistency: Choose $\eta_n \asymp (q_n \log n)^{\frac{1}{d+2}}$ and $\delta_n \asymp \sqrt{\eta_n + e_n}$,

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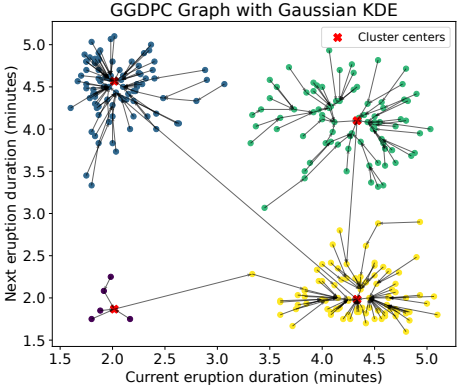
Criterion	Rate	Interpretation
Mode set (Hausdorff)	$O_P\left(r_n^{\min\left\{\frac{2}{d+6}, \frac{1}{d}\right\}}\right)$	1NN error dominates for $d \geq 7$
Clustering ($1 - \text{ARI}$)	$O_P\left(r_n^{\min\left\{\frac{2}{d+6}, \frac{1}{2d}\right\}}\right)$	1NN error dominates for $d \geq 3$
Dendrogram (d_{GH})	$O_P\left((\log \log n) r_n^{\min\left\{\frac{2}{d+6}, \frac{1}{2d}\right\}}\right)$	Same polynomial rate as ARI
Path length	$O_P\left(\left[(\log n)^{\frac{1}{d+2}} r_n^{\frac{1}{d(d+2)}} + r_n^{\frac{2}{d+6}}\right]^{\frac{1}{2}}\right)$	1NN error dominates for $d \geq 2$

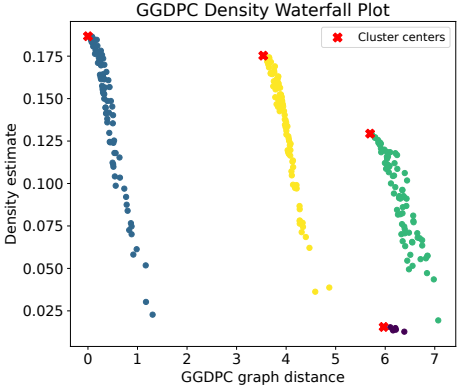
Density Waterfall Plot

- $\boldsymbol{X}^* = \arg\max_{\boldsymbol{X}_j \in \mathbb{X}_n} \widehat{p}(\boldsymbol{X}_j)$ is the sample global mode.
- $\boldsymbol{X}_i \equiv \boldsymbol{X}_{i_1} \to \boldsymbol{X}_{i_2} \to \cdots \to \boldsymbol{X}_{i_{T_i}} \equiv \boldsymbol{X}^*$: unique directed path in the GGDPC graph G.
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Density Waterfall Plot: $\left\{ \left(\widehat{d}_G(\boldsymbol{X}_i), \widehat{p}(\boldsymbol{X}_i) \right) : i = 1, ..., n \right\}$.

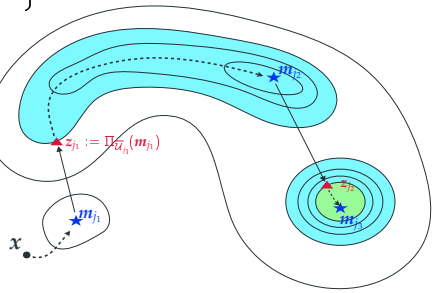




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- Empirical density waterfall measure: $\forall A \in \text{Borel}(\mathbb{R}^2)$,

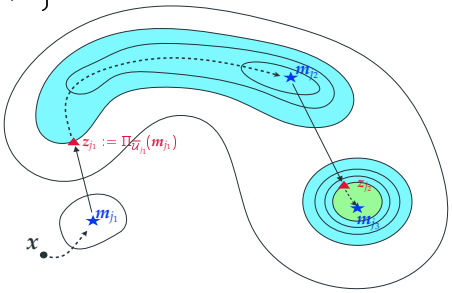
$$\widehat{Q}_n(A) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1} \left\{ \left(\widehat{d}_G(\boldsymbol{X}_i), \widehat{p}(\boldsymbol{X}_i) \right) \in A \right\}.$$



Density Waterfall Plot: $\left\{ \left(\widehat{d}_G(\boldsymbol{X}_i), \widehat{p}(\boldsymbol{X}_i) \right) : i = 1, ..., n \right\}$.

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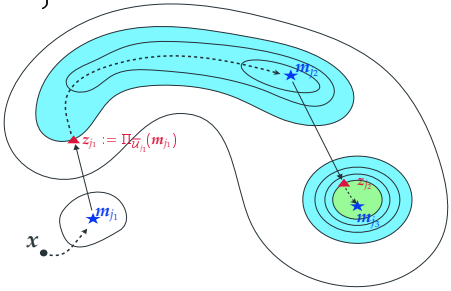


- Population GGDPC graph distance:

$$\bar{d}_G(\boldsymbol{x}) = L(\boldsymbol{x}) + \sum_{\ell=1}^{\bar{T}_{\boldsymbol{x}}-1} \left[\left\| \boldsymbol{m}_{j_\ell} - \Pi_{\overline{\mathcal{U}}_{j_\ell}}(\boldsymbol{m}_{j_\ell}) \right\| + L\left(\Pi_{\overline{\mathcal{U}}_{j_\ell}}(\boldsymbol{m}_{j_\ell}) \right) \right].$$

- Population density waterfall measure: $Q(A) = \mathbb{P}\left(\left(\bar{d}_G(\boldsymbol{X}), p(\boldsymbol{X}) \right) \in A \right)$.

Density Waterfall Plot: $\left\{ \left(\widehat{d}_G(\boldsymbol{X}_i), \widehat{p}(\boldsymbol{X}_i) \right) : i = 1, ..., n \right\}$.



- Empirical density waterfall measure: $\forall A \in \text{Borel}(\mathbb{R}^2)$,

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- Population GGDPC graph distance:

$$\bar{d}_G(\boldsymbol{x}) = L(\boldsymbol{x}) + \sum_{\ell=1}^{\bar{T}_{\boldsymbol{x}}-1} \left[\left|\left| \boldsymbol{m}_{j_\ell} - \Pi_{\overline{\mathcal{U}}_{j_\ell}}(\boldsymbol{m}_{j_\ell}) \right|\right| + L\left(\Pi_{\overline{\mathcal{U}}_{j_\ell}}(\boldsymbol{m}_{j_\ell}) \right) \right].$$

- Population density waterfall measure: $Q(A) = \mathbb{P}\left(\left(\bar{d}_G(\boldsymbol{X}), p(\boldsymbol{X}) \right) \in A \right)$.

$$\text{Wass}_1(\widehat{Q}_n, Q) = O_P\left(\text{“Path-Length Consistency Rate”} + \delta_n + ||\widehat{p} - p||_\infty + \frac{\log n}{\sqrt{n}} \right).$$

δ_n: observations fell into the boundary tube; $\left|\left|\widehat{p} - p\right|\right|_\infty$: density estimation error; $\log n/\sqrt{n}$: Wasserstein-1 rate.

Empirical Studies and Conclusion



Comparisons With Other DPC-Type Methods on a Two-Gaussian Mixture

Sample size	500	1000	2000	5000	8000
Adjusted Rand index					
DPC (Rodriguez and Laio, 2014)	.898	.920	.935	.946	.953
DPC-KNN-PCA (Du et al., 2016)	**.935**	**.939**	.943	.945	.948
SNN-DPC (Liu et al., 2018)	.881	.871	.872	.874	.878
DPC-CE (Guo et al., 2022)	.840	.886	.911	.928	.941
DPC-DLP (Seyedi et al., 2019)	.341	.494	.632	.772	.808
DPC-MDNN (Wang et al., 2025)	.211	.077	.010	.000	.000
GGDPC (Ours)	.916	.938	**.953**	**.963**	**.967**
Running time (seconds)					
DPC (Rodriguez and Laio, 2014)	.039	.100	.347	1.858	4.627
DPC-KNN-PCA (Du et al., 2016)	**.020**	.075	.313	2.239	6.096
SNN-DPC (Liu et al., 2018)	.167	1.111	9.922	159.913	632.394
DPC-CE (Guo et al., 2022)	.728	1.802	4.896	20.686	45.763
DPC-DLP (Seyedi et al., 2019)	.074	.337	1.634	14.996	41.864
DPC-MDNN (Wang et al., 2025)	.262	1.128	5.770	56.383	180.123
GGDPC (Ours)	.028	**.067**	**.203**	**1.061**	**2.601**

Average over 1,000 replications. GGDPC times exclude the optional dendrogram.

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Local Mode Recovery | Adjusted Rand Index | Dendrogram | Path Length | Waterfall Measure

Local — **Geometric concepts** — Global



Fast — **Convergence Rate** — Slow

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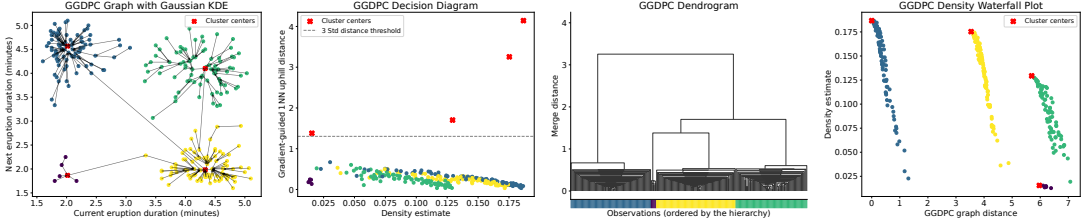
Theory Five criteria for its consistency:

Local Mode Recovery | Adjusted Rand Index | Dendrogram | Path Length | Waterfall Measure

Local — Geometric concepts — Global

Fast — Convergence Rate — Slow

Infographic GGDPC graph, decision diagram, modal distance dendrogram, density waterfall.



Future Directions

1. **Improved convergence rate** via direct mapping within the modal region $B\left(\widehat{\boldsymbol{m}}_j, \epsilon_n\right)$,

$$\text{“path-length consistency”} = O_P\left(\left\{\left[\frac{(\log n)^{d+1}}{n}\right]^{\frac{1}{d(d+2)}} + \left(\frac{\log n}{n}\right)^{\frac{2}{d+6}}\right\}^{\frac{1}{2}}\right) \Longrightarrow O_P\left(\left(\frac{\log n}{n}\right)^{\frac{1}{3d}} \log n\right).$$

2. **Local signatures via density waterfall plots** for mean shift or k-means clustering.

1. **Improved convergence rate** via direct mapping within the modal region $B\left(\widehat{\boldsymbol{m}}_j, \epsilon_n\right)$,

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2. **Local signatures via density waterfall plots** for mean shift or k-means clustering.

Thank you!

Paper: https://arxiv.org/abs/2610.01050.

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Precise Regularity Conditions

A1: Density. (a) p has a compact support $\mathcal{C}$ with $\inf_{x\in\mathcal{C}} p(\boldsymbol{x}) \geq p_{\min} > 0$;
(b) $\exists C_{\mathcal{C}}, \epsilon_0 > 0$ s.t. $\text{Leb}\left(\mathcal{C} \cap B(\boldsymbol{x}, \epsilon)\right) \geq C_{\mathcal{C}} \cdot \text{Leb}\left(B(\boldsymbol{x}, \epsilon)\right), \forall \boldsymbol{x} \in \mathcal{C}$ and $0 < \epsilon \leq \epsilon_0$;
(c) p has a C^3 extension to a neighborhood with bounded derivatives;
(d) Its critical points are finite and non-degenerate; distinct modal heights;
(e) $\mathcal{C}$ is forward-invariant for $\boldsymbol{\gamma}_{\boldsymbol{x}}'(t) = \nabla p(\boldsymbol{\gamma}_{\boldsymbol{x}}(t))$.

A2: Estimation. On a fixed neighborhood of $\mathcal{C}$, $\widehat{p} \in C^2$, $\widehat{g} = \nabla\widehat{p}$, and derivatives through order two are bounded in probability.

A3: Separatrix. (a) Each $\partial\mathcal{C}_a$ is a finite union of stable manifolds of saddles with one positive and $d-1$ negative Hessian eigenvalues;
(b) It is a C^2 hypersurface away from finitely many critical points. Write $\mathcal{S}_{a,\text{reg}}$ for its noncritical C^2 part. For some $r_{\mathcal{S}}, \rho_{\mathcal{S}} > 0$:

$$\boldsymbol{x} \in \mathcal{C}_a, \quad 0 < d(\boldsymbol{x}, \partial\mathcal{C}_a) < r_{\mathcal{S}}, \quad \Pi_{\partial\mathcal{C}_a}(\boldsymbol{x}) \cap \mathcal{S}_{a,\text{reg}} \neq \varnothing \implies |\Pi_{\partial\mathcal{C}_a}(\boldsymbol{x})| = 1,$$

$$\nu(\boldsymbol{s})^T \nabla^2 p(\boldsymbol{s}) \nu(\boldsymbol{s}) \geq \rho_{\mathcal{S}} \quad \text{on } \mathcal{S}_{a,\text{reg}} \text{ outside fixed saddle neighborhoods.}$$

A4: Projections. For every non-global mode $\boldsymbol{m}_j$ ($j \geq 2$), the projection $\boldsymbol{z}_j = \Pi_{\overline{\mathcal{U}}_j}(\boldsymbol{m}_j)$ is unique, $\boldsymbol{z}_j \in \mathcal{C}_{\pi(j)}$, $\nabla p(\boldsymbol{z}_j) \neq 0$, and $d(\boldsymbol{z}_j, \partial\mathcal{C}_{\pi(j)}) > 0$. For some $\kappa_j > 0$,

$$\boldsymbol{v}^T[I_d - \mu_j \nabla^2 p(\boldsymbol{z}_j)]\boldsymbol{v} \geq \kappa_j \quad \text{if } \|\boldsymbol{v}\| = 1,\ \boldsymbol{v} \perp \nabla p(\boldsymbol{z}_j), \qquad \boldsymbol{z}_j - \boldsymbol{m}_j = \mu_j \nabla p(\boldsymbol{z}_j), \quad \mu_j > 0.$$

A5: Saddles. ∇p is C^1-linearizable near every boundary saddle.

Stable and unstable manifolds: Let s be a critical point of p, and $\gamma_x'(t) = \nabla p(\gamma_x(t))$:

$$W^s(s) = \{x \in C : \lim_{t\to+\infty} \gamma_x(t) = s\}, \qquad W^u(s) = \{x \in C : \lim_{t\to-\infty} \gamma_x(t) = s\}.$$

- By the stable manifold theorem, $W^s(s), W^u(s)$ are locally C^2 manifolds in $\mathbb{R}^d$ near s.

C^1-linearization: There exists a local C^1-diffeomorphism Φ with $\Phi(\mathbf{0}) = \mathbf{0}$ satisfying

$$\Phi(\gamma_x(t)) = e^{tA}\Phi(x), \qquad A = \nabla^2 p(s),$$

while trajectories stay in the local charts. In these coordinates, $y'(t) = Ay(t)$.

Sternberg non-resonance condition:

For eigenvalues $\rho_1, \ldots, \rho_d$ of $\nabla^2 p(s)$, require at every order $N \geq 2$,

$$\rho_i - \sum_{j=1}^{d} k_j \rho_j \neq 0, \quad i = 1, \ldots, d,$$

$$k_j \in \mathbb{N}, \quad 2 \leq \sum_{j=1}^{d} k_j \leq N.$$

Hartman spectral condition:

The eigenvalues of $\nabla^2 p(s)$ lie in

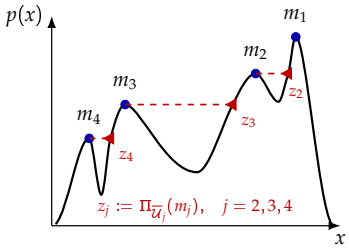
$$[-\alpha_L, -\alpha_R], \quad [\beta_L, \beta_R], \quad \text{with } \alpha_R, \alpha_L, \beta_L, \beta_R > 0.$$

Require $\alpha_L - \alpha_R < \mu\beta_L$ and $\beta_R - \beta_L < \nu\alpha_R$,

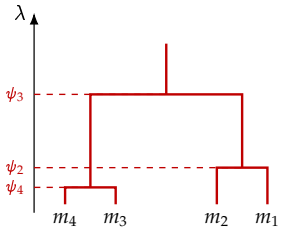
- $\mu, \nu > 0$ are the Hölder spectral exponents of the stable/unstable linearized flow.

Distinctions Between Modal Distance Dendrogram and Density Cluster Tree

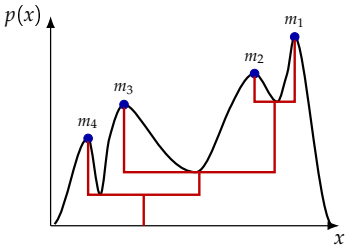
The modal distance dendrogram $\mathcal{T}_{\text{mode}}$ and the density cluster tree $\mathcal{T}_p$ can have different merge topologies.



(a) Density p and modal projections



(b) Modal-distance dendrogram $\mathcal{T}_{\text{mode}}$



(c) Density cluster tree $\mathcal{T}_p$

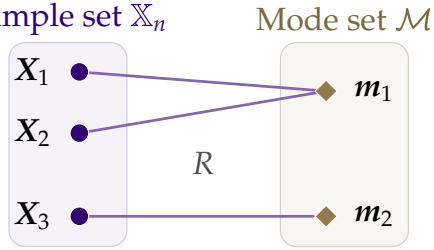
- Modal distance tree: $w(\boldsymbol{m}_4) < w(\boldsymbol{m}_2) \ll w(\boldsymbol{m}_3)$, so close pairs merge first.
- Density tree: merges follow the ordered density levels.

An *ultrametric* on a set $\mathbb{X}$ is a function $u : \mathbb{X} \times \mathbb{X} \to \mathbb{R}_+$ satisfying, for any $x, x', x'' \in \mathbb{X}$,

(a) $u(x, x') = 0 \Leftrightarrow x = x'$; (b) $u(x, x') = u(x', x)$; (c) $u(x, x') \leq \max\{u(x, x''), u(x'', x')\}$.

- $u_n(X_i, X_j) = \max\left\{\widehat{w}_n(X_k) : \left(X_k \to \widehat{\Phi}_n(X_k)\right) \in \mathrm{Path}_G(X_i, X_j)\right\}$, where $\mathrm{Path}_G(X_i, X_j)$ is the edge set in G joining X_i and X_j.
- $u_{\mathcal{M}}(m_j, m_k) = \max\left\{w(m_\ell) : \left(m_\ell \to m_{\pi(\ell)}\right) \in \mathrm{Path}_{G_{\mathcal{M}}}(m_j, m_k)\right\}$.

A *correspondence* $R \subseteq \mathbb{X}_n \times \mathcal{M}$ satisfies both coordinate projections are surjective.



The **Gromov-Hausdorff distance** between $(\mathbb{X}_n, u_n)$ and $(\mathcal{M}, u_{\mathcal{M}})$ is defined as

$$d_{\mathrm{GH}}((\mathbb{X}_n, u_n), (\mathcal{M}, u_{\mathcal{M}})) = \frac{1}{2} \inf_R \sup_{(x,m),(y,m') \in R} \left| u_n(x, y) - u_{\mathcal{M}}(m, m') \right|.$$

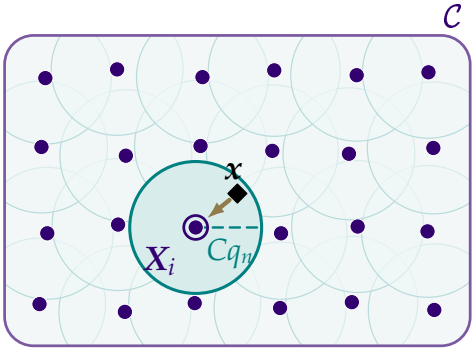
Uniform Sample Coverage Lemma

Lemma (Lemma C.1 in Zhang and Chen 2026)

Suppose that Assumption A1(a) holds, and the density p is bounded on its support $\mathcal{C}$. Then, almost surely,

$$\sup_{\boldsymbol{x}\in\mathcal{C}} \min_{i=1,\ldots,n} ||\boldsymbol{x} - \boldsymbol{X}_i|| = O\left(\left(\frac{\log n}{n}\right)^{\frac{1}{d}}\right).$$

The same bound holds over any closed subset of $\mathcal{C}$.



Purple dots are samples; the gold arrow joins $\boldsymbol{x}$ to its nearest sample.

As $n \to \infty$, almost surely, sample neighborhoods cover the support: $\mathcal{C} \subseteq \bigcup_{i=1}^{n} B(\boldsymbol{X}_i, Cq_n)$.